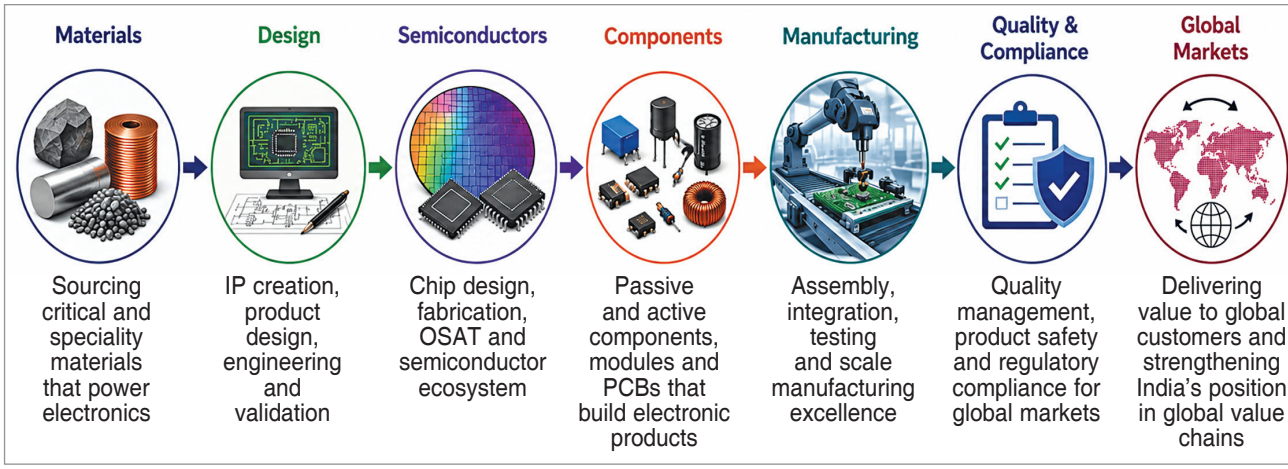


India's electronics value chain

patents and technologies originating in India. Partnerships with multinational companies remain important for technology access, investment and global markets. The objective is to increase the share of innovation that is originated, owned and commercialised from India.

Advanced packaging is another emerging frontier. In August 2026, the Centre for Materials for Electronics Technology (C-MET) Pune invited Indian industry participation in developing advanced photonics packaging technologies using additive manufacturing for photonic integrated circuits. As computing becomes more powerful, packaging and interconnect technologies are increasingly important for moving data efficiently between chips and systems, including AI data-centre interconnects.

India's semiconductor opportunity therefore extends beyond making chips to designing, packaging, connecting, testing and integrating them into sophisticated systems.

Self-reliance without isolation

Technological self-reliance should not mean reproducing every part of a global value chain domestically. A sophisticated electronic product may contain intellectual property from one country, wafers fabricated in another, packaging elsewhere, materials from several regions and software developed across multiple locations.

A stronger definition is the ability to participate in global technology networks from a position of capability rather than excessive dependence. That requires domestic strength in strategically important technologies, competitive Indian suppliers, multiple supply options, international partnerships, engineering capability and resilient supply chains.

Bharat Mandapam and Yashobhoomi in New Delhi increasingly provide platforms where global industry, technology, innovation and investment converge. SEMICON India 2026 was held from September 17 to 19 at Yashobhoomi, bringing together the global semiconductor ecosystem. The event underscored the transition from semiconductor ambition to a broader silicon-to-systems ecosystem.

Self-reliance and global integration are therefore not opposites. A technologically stronger India can become a more valuable participant in global electronics supply chains.

Can quality keep pace with scale?

Production figures make headlines; reliability determines whether customers return. Defective components can stop assembly lines, weak process control can create large-scale failures, and poor traceability can turn a contained defect into an expensive recall.

As Indian electronics manufacturing expands, quality infrastructure must expand with it. Supplier quality, component qualification, reliability testing, failure analysis, process control, calibration, product safety, traceability and corrective-action systems become increasingly important, particularly for suppliers entering multinational supply chains.

Competitiveness will depend not only on price, but on consistent performance, reliable data, traceable processes and evidence that problems can be identified and corrected. For Indian electronics suppliers, quality is not merely a factory function; it is market access.

Compliance becomes competitiveness

The same principle applies to environmental and regulatory compliance. As Indian electronics exports expand, products encounter requirements such as the European Union's Restriction of Hazardous Substances (RoHS) and Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), persistent organic pollutants controls, Per- and polyfluoroalkyl substances (PFAS) requirements, battery and packaging rules, responsible-minerals reporting, ecodesign, repairability,